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QUESTIONS

Q1 Existing System Analysis
20 marks**Q2** Literature Survey
20 marks**Q3** System Design
20 marks**Q4** Project Planning
20 marks**Q5** Individual Module Presentation
20 marks

5/5 answered

Q1 Existing System Analysis

20 marks

Analyze the existing system related to your project. Explain the limitations, challenges, and justify the need for your proposed solution. Include at least three limitations of the existing system.

Your Answer

Existing System

Currently, space object tracking and collision risk assessment rely mainly on systems like NASA's Orbital Debris Program Office, ESA's Space Debris Office, and Space-Track.org, which provide TLE (Two-Line Element) data and periodic conjunction assessment reports. Organizations and researchers typically access this data through static websites, downloadable text files, or specialized (often expensive) software like STK (Systems Tool Kit) or AGI's tools.

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Q2 Literature Survey

20 marks

Review at least five research papers related to your project. Prepare a comparison table showing the title, methodology, advantages, limitations, and research gap.

Your Answer

Title	Methodology	Advantages	Limitations	Research Gap
Revisiting Spacetrack Report #3: Revitalizing SGP4 Orbit Determination (Vallado et al., 2006)	Re-derives and standardizes the SGP4/SDP4 orbital propagation algorithms from TLE data	Provides the mathematical foundation for accurate, standardized orbit propagation; widely adopted industry standard	Purely mathematical/algorithmic — no visualization or real-time monitoring interface	Does not address how to present propagated data to end users in an accessible, real-time way
Celestrak: Current GP Element Sets (Kelso, 2024)	Aggregates and publishes live TLE data for satellites and debris from public			

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5/5 answered

Q3 System Design

20 marks

Present the design of your proposed system by including any three of the following: System Architecture, Datasets

Your Answer

1. System Architecture

OrbitGuard follows a 3-tier architecture:

Data Source Layer – CelesTrak provides live TLE (Two-Line Element) data for active satellites (e.g., ISS, Starlink) and space debris (e.g., Cosmos-2251, Iridium-33).

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5/5 answered

Q4 Project Planning

20 marks

Present your project implementation plan including module breakdown, Gantt chart, technology stack, expected timeline, and future work. Demonstrate your initial prototype (if available) and answer the review committee's questions.

Your Answer**Module Breakdown**

Data Ingestion Module – Fetches live TLE data from CelesTrak at regular intervals.
Orbital Propagation Module – Uses Skyfield/SGP4 to calculate position, velocity, altitude, and heading for each object.
Risk Assessment Module – Computes pairwise distances and assigns HIGH/MEDIUM/LOW risk levels.
API Service Module – Spring Boot REST API exposing processed data as JSON.

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5/5 answered

Q5 Individual Module Presentation

20 marks

Present your individual module and its flow and model output20

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